

Lab 1: OLS by Hand, in Scalar Form

SOCIOL 723, Social Statistics II, Thursday, August 27

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1 Goals

By the end of this lab you should be able to:

1. Organize an analysis in RStudio and Quarto so that it is reproducible.
2. Compute $\hat{\beta}_0$ and $\hat{\beta}_1$ from the scalar formulas with nothing but `sum()`, and verify that they match `lm()`.
3. Verify the mechanical residual properties, and compute s^2 and the classical standard error by hand.
4. Use **simulation** to see what the assumption $E[\epsilon_i | X_i] = 0$ does and does not deliver.
5. Verify the Frisch–Waugh–Lovell theorem and the omitted-variable-bias formula in real GSS data.

Everything below is scalar arithmetic: sums, means, and `lm()`. No matrix computations appear anywhere in this lab.

2 Getting set up

2.1 Packages

```
library(dplyr)      # data manipulation
library(ggplot2)    # graphics
library(broom)      # tidy() / augment() for model objects
library(modelsummary) # regression tables
```

If any of these are missing, install them once with `install.packages(c("dplyr", "ggplot2", "broom", "modelsummary"))`.

2.2 Working reproducibly

Three habits, which I will assume for every problem set:

- **Use an RStudio Project.** File > New Project. Never use `setwd()`; paths should be relative to the project root.
- **Write in Quarto (.qmd) or RMarkdown (.Rmd).** Your submission should be a rendered PDF produced from a source file that I could run myself.
- **Set a seed** before anything random: `set.seed(1)`.

A useful discipline: restart R (`Session > Restart R`, or `Cmd/Ctrl+Shift+F10`) and re-render from a clean session before you submit. If it does not run from scratch, it is not reproducible.

3 The data

We use an extract from the **General Social Survey** (NORC) cumulative file, pooling the 2010, 2012, 2014, 2016, 2018 and 2022 waves. The sample is restricted to respondents working full time, aged 25–64, with positive personal income and non-missing values on the analysis variables.

```

gss <- readRDS("../Data/gss_earnings.rds")
dim(gss)
#> [1] 3509    23
str(gss, give.attr = FALSE)
#> 'data.frame':    3509 obs. of  23 variables:
#> $ year      : int  2010 2010 2010 2010 2010 2010 2010 2010 2010 2010 2010 ...
#> $ id        : int   1 13 20 21 29 35 38 42 44 50 ...
#> $ llearn    : num  10.66 9.87 10.66 10.26 10.46 ...
#> $ realrinc  : num  42735 19425 42735 28490 34965 ...
#> $ educ      : num   16 12 16 12 16 15 16 18 17 12 ...
#> $ degree    : Ord.factor w/ 5 levels "< HS"<"HS"<"Junior college"<...: 4 2 4 2 4 2 5 5 5 2 .
#> $ ba        : int   1 0 1 0 1 0 1 1 1 0 ...
#> $ exper     : num   9 40 13 31 11 36 31 9 2 22 ...
#> $ age       : num  31 58 35 49 33 57 53 33 25 40 ...
#> $ female    : int   0 0 0 0 1 0 0 0 0 1 ...
#> $ black     : int   0 1 0 0 0 0 0 0 0 0 ...
#> $ otherace  : int   1 0 0 0 0 0 0 1 1 1 ...
#> $ wordsum   : num   6 5 6 6 8 7 3 8 3 2 ...
#> $ pareduc   : num   8 16 12 12 12 12 18 12 18 6 ...
#> $ prestige  : num  60 24 33 49 47 41 41 61 46 30 ...
#> $ sei       : num  76.3 20.7 46.2 45.4 38.8 54.6 54.6 80.9 61.4 20.1 ...
#> $ hours     : num  55 40 50 40 40 60 40 35 60 50 ...
#> $ childs    : num   0 2 0 0 0 2 2 0 0 2 ...
#> $ evermar   : int   0 1 0 1 0 1 1 0 0 1 ...
#> $ region    : Factor w/ 9 levels "New England",...: 2 2 2 2 2 2 2 2 2 1 ...
#> $ stratum   : int  2240 2242 2243 2243 2244 2245 2245 2246 2246 2247 ...
#> $ psu       : int  22401 22421 22431 22431 22442 22451 22452 22461 22461 22471 ...
#> $ wt        : num   1.127 0.64 0.447 0.554 0.39 ...

```

Variable	Meaning
<code>llearn</code>	log of personal income, constant dollars (<code>realrinc</code>)
<code>educ</code>	years of schooling completed
<code>exper</code>	potential experience, $\max(\text{age} - \text{educ} - 6, 0)$
<code>female</code> , <code>black</code> , <code>otherace</code>	indicator variables
<code>wordsum</code>	score on a 10-word vocabulary test (proxy for verbal ability)
<code>pareduc</code>	highest education of either parent, in years
<code>prestige</code> , <code>sei</code>	occupational prestige and socioeconomic index
<code>hours</code>	usual hours worked last week
<code>region</code>	Census division (9 categories)
<code>stratum</code> , <code>psu</code>	GSS sampling stratum and primary sampling unit (used in Lab 2)
<code>wt</code>	post-stratification sampling weight

i Note

The script that builds this file from the raw cumulative file is in `Data/build_gss_extract.R`. You do not need to run it.

A first look:

```
gss |>
  summarise(across(c(llearn, educ, exper, wordsum, pareduc, hours),
                    list(mean = mean, sd = sd), .names = "{.col}_{.fn}")) |>
  t() |> round(2)
#>           [,1]
#> llearn_mean  9.95
#> llearn_sd    0.93
#> educ_mean   14.48
#> educ_sd     2.82
#> exper_mean  23.18
#> exper_sd    11.22
#> wordsum_mean 6.33
#> wordsum_sd   1.91
#> pareduc_mean 13.20
#> pareduc_sd   3.61
#> hours_mean  45.49
#> hours_sd    11.33
```

```
gss |>
  group_by(educ) |>
  summarise(m = mean(llearn), n = n()) |>
  filter(n >= 20) |>
  ggplot(aes(educ, m)) +
  geom_point(aes(size = n), colour = "navy", alpha = .7) +
  geom_smooth(method = "lm", se = FALSE, colour = "firebrick", linewidth = .6) +
  labs(x = "Years of schooling", y = "Mean log earnings", size = "n") +
  theme_minimal(base_size = 10)
```

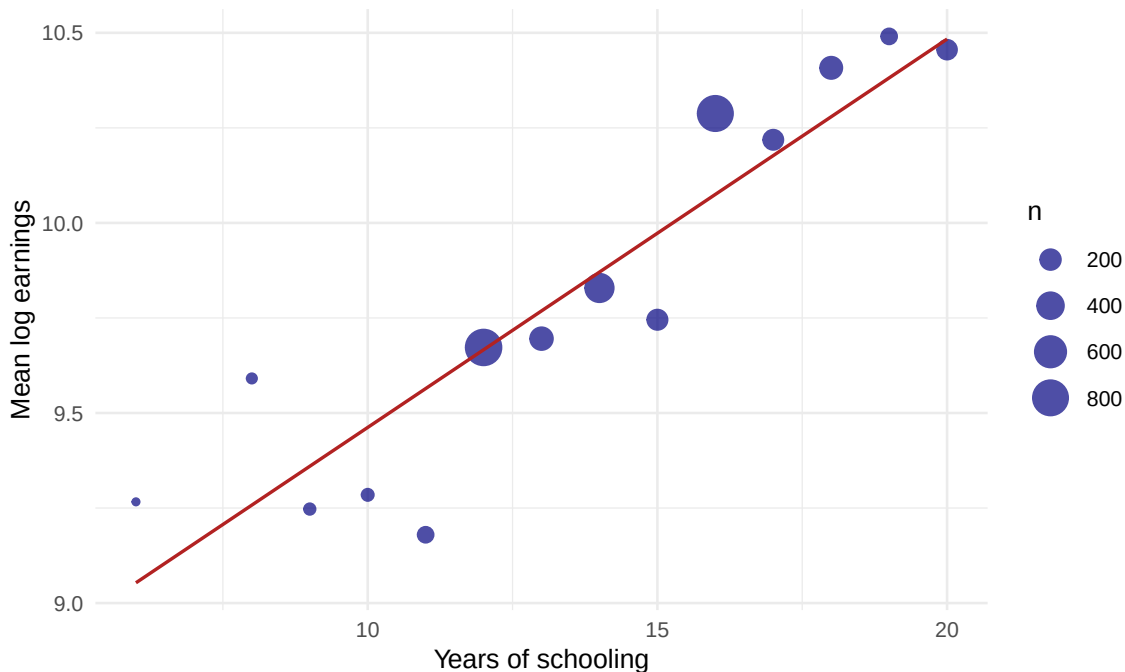


Figure 1: The CEF of log earnings given years of schooling, GSS 2010–2022.

The points are the **conditional expectation function** estimated cell by cell (this is what a *saturated* regression would fit). The line is the **best linear approximation** to it. Notice that the CEF is not exactly linear; it is convex above roughly 12 years, but the line summarizes it reasonably well.

4 OLS by hand, in scalar form

4.1 The slope and intercept from sums

The whole estimator is two lines of arithmetic: $\hat{\beta}_1 = \text{Cov}_n(X, Y) / \text{Var}_n(X)$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$.

```
x <- gss$educ
y <- gss$lnearn
n <- length(y)

b1 <- sum((x - mean(x)) * (y - mean(y))) / sum((x - mean(x))^2)
b0 <- mean(y) - b1 * mean(x)
round(c(b0 = b0, b1 = b1), 4)
#>      b0      b1
#> 8.2307 0.1187
```

Compare with `lm()`:

```
fit_biv <- lm(lnearn ~ educ, data = gss)
round(coef(fit_biv), 4)
```

```
#> (Intercept)      educ
#>      8.2307      0.1187
```

Identical, and identical to the numbers on the lecture slides. `lm()` is doing exactly this arithmetic (in a numerically fancier way), nothing more.

4.2 The slope is a weighted average of unit slopes

Lecture claimed $\hat{\beta}_1 = \sum_i w_i (Y_i - \bar{Y}) / (X_i - \bar{X})$ with $w_i = (X_i - \bar{X})^2 / \sum_j (X_j - \bar{X})^2$. Verify it, and look at who carries the weight:

```
w <- (x - mean(x))^2 / sum((x - mean(x))^2)
bi <- ifelse(x == mean(x), 0, (y - mean(y)) / (x - mean(x)))
c(weighted_average = sum(w * bi), b1 = b1)
#> weighted_average      b1
#>      0.1187      0.1187

## total weight carried by each value of educ
tapply(w, x, sum) |> round(3)
#>      0      2      3      4      5      6      7      8      9     10     11     12     13
#> 0.022 0.017 0.019 0.012 0.010 0.054 0.020 0.045 0.042 0.032 0.041 0.183 0.020
#>     14     15     16     17     18     19     20
#> 0.004 0.002 0.066 0.042 0.108 0.069 0.193
```

Respondents far from the mean of schooling (about 14.5 years) carry almost all of the weight; respondents at the mean carry essentially none. Know which observations are shouting.

4.3 Residual mechanics

```
e <- y - b0 - b1 * x
round(c(sum_e = sum(e), sum_xe = sum(x * e), cor_xe = cor(x, e)), 8)
#> sum_e sum_xe cor_xe
#>      0      0      0
```

These hold to machine precision in *every* sample, no matter how wrong the model is. They are consequences of the normal equations, not evidence of anything.

4.4 s^2 and the classical standard error

```
s2 <- sum(e^2) / (n - 2)
se_b1 <- sqrt(s2 / sum((x - mean(x))^2))
round(c(s2 = s2, se_b1 = se_b1), 5)
#>      s2      se_b1
#> 0.76066 0.00521
round(summary(fit_biv)$coefficients["educ", ], 5)
```

```
#>      Estimate Std. Error      t value    Pr(>|t|)
#>      0.11873      0.00521     22.77854      0.00000
```

The standard error R prints is exactly $\sqrt{s^2 / \sum_i (X_i - \bar{X})^2}$, i.e. it *assumes homoskedasticity*. Next week we replace it.

5 Simulation: what the error assumption does

With real data we never know the truth. In a simulation we do, which makes it the honest way to see what an assumption buys.

5.1 When the average of Y really is a line

```
set.seed(1)
n_sim <- 2000
xs <- sample(8:20, n_sim, replace = TRUE)      # schooling-like regressor
ys <- 8.2 + 0.12 * xs + rnorm(n_sim, 0, 0.7)    # E[eps | x] = 0 by design
coef(lm(ys ~ xs)) |> round(4)
#> (Intercept)          xs
#>      8.1642      0.1235
```

We wrote the truth ($\beta_0 = 8.2$, $\beta_1 = 0.12$) and OLS finds it, up to sampling noise. Rerun with a different seed: the estimate moves a little. How much it moves from sample to sample is precisely next week's topic.

5.2 When the CEF is not a line: uncorrelated is not mean-independent

Now make the population relationship curved, and fit a line anyway:

```
set.seed(2)
xq <- runif(n_sim, 8, 20)
yq <- 8 + 0.05 * xq + 0.015 * (xq - 14)^2 + rnorm(n_sim, 0, 0.4)
fq <- lm(yq ~ xq)

round(cor(xq, resid(fq)), 8)      # exactly zero, mechanically
#> [1] 0

plot(xq, resid(fq), pch = 16, cex = .35, col = "navy",
     xlab = "x", ylab = "residual")
lines(lowess(xq, resid(fq)), col = "firebrick", lwd = 2)
abline(h = 0, lty = 2)
```

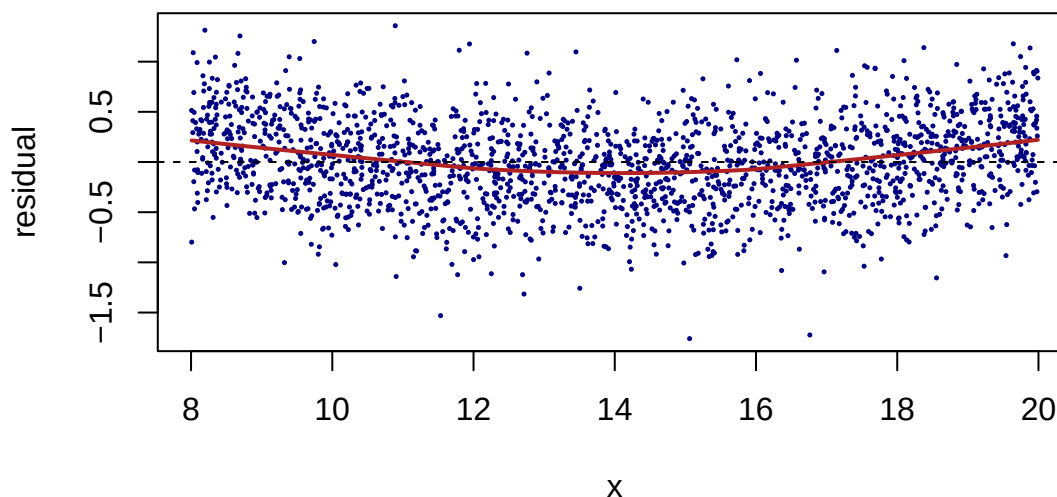


Figure 2: Residuals from fitting a line to a curved CEF. Their correlation with x is exactly zero, yet their average tracks the leftover curve: uncorrelated, but not mean-independent.

This is the ladder from lecture, live: the residuals are *uncorrelated* with x (bottom rung, guaranteed by the algebra) but their *average* clearly moves with x , so mean independence fails. The correlation could never have told you; the plot does. This is why you look at residual plots.

5.3 Heteroskedasticity does not bias the slope

```
set.seed(3)
slopes <- replicate(500, {
  xh <- runif(500, 8, 20)
  yh <- 8.2 + 0.12 * xh + rnorm(500, 0, 0.1 * xh) # spread grows with x
  coef(lm(yh ~ xh))[2]
})
round(c(mean_slope = mean(slopes), truth = 0.12), 4)
#> mean_slope      truth
#>    0.1209      0.1200
```

The error spread rises steeply with x , yet the slope estimates still centre on the truth: heteroskedasticity is a problem for *standard errors* (Week 2), not for the estimate itself. $E[\epsilon | X] = 0$ still holds here, and that is the assumption doing the work.

6 The multiple regression, and fit

With several regressors the same logic goes through; we let `lm()` carry the bookkeeping. (Thursday's session shows the notation that makes this compact.)

```
fit <- lm(llearn ~ educ + exper + I(exper^2) + female + black, data = gss)
y <- gss$llearn
round(coef(fit), 4)
```



```
#> (Intercept)      educ      exper  I(exper^2)      female      black
#>      7.3901      0.1405      0.0529      -0.0007     -0.3758     -0.1915
```

6.1 The sum-of-squares decomposition

```
yhat <- fitted(fit)
TSS <- sum((y - mean(y))^2)
ESS <- sum((yhat - mean(y))^2)
RSS <- sum(residuals(fit)^2)

c(TSS = TSS, ESS_plus_RSS = ESS + RSS)
#>      TSS ESS_plus_RSS
#>    3062         3062
c(R2_manual = ESS / TSS, R2_lm = summary(fit)$r.squared)
#> R2_manual      R2_lm
#>   0.2326     0.2326
```

And the model-comparison reading of R^2 : it is *PRE* against the intercept-only model.

```
fit_c <- lm(lnearn ~ 1, data = gss) # MODEL C
SSE_C <- sum(residuals(fit_c)^2)
SSE_A <- sum(residuals(fit)^2) # MODEL A
c(PRE = (SSE_C - SSE_A) / SSE_C, R2 = summary(fit)$r.squared)
#>  PRE      R2
#> 0.2326 0.2326
```

7 Frisch–Waugh–Lovell

We now verify FWL on the return to schooling: the treatment of interest is `educ`, and the controls are everything else in the regression.

```
## Step 1: residualize the treatment on the controls
d_tilde <- residuals(lm(educ ~ exper + I(exper^2) + female + black, data = gss))

## Step 2: residualize the outcome on the controls
y_tilde <- residuals(lm(lnearn ~ exper + I(exper^2) + female + black, data = gss))

## Step 3: bivariate regression of the residuals
b_fwl <- coef(lm(y_tilde ~ d_tilde))["d_tilde"]

c(full_regression = coef(fit)["educ"], fwl = b_fwl)
#> full_regression.educ      fwl.d_tilde
#>          0.1405          0.1405
```

Both forms of the regression-anatomy formula:

```
c(cov_ytilde_dtilde = cov(y_tilde, d_tilde) / var(d_tilde),
  cov_y_dtilde      = cov(gss$lnearn, d_tilde) / var(d_tilde))
#> cov_ytilde_dtilde      cov_y_dtilde
#>          0.1405          0.1405
```

In fact *all three* numerators agree, for the same reason:

```
c(both_residualized = cov(y_tilde, d_tilde) / var(d_tilde),
  only_regressor     = cov(gss$lnearn, d_tilde) / var(d_tilde),
  only_outcome       = cov(y_tilde, gss$educ) / var(d_tilde))
#> both_residualized      only_regressor      only_outcome
#>          0.1405          0.1405          0.1405
```

What you *cannot* change is the **denominator**:

```
c(correct = coef(lm(y_tilde ~ d_tilde))["d_tilde"],
  wrong   = coef(lm(y_tilde ~ gss$educ))[2])
#> correct.d_tilde      wrong.gss$educ
#>          0.1405          0.1293
```

i Note

Why does the second one fail? Regressing $\tilde{Y}$ on raw `educ` divides by $\text{Var}_n(\text{educ})$ instead of $\text{Var}_n(\widetilde{\text{educ}})$. Since residualizing can only shrink the variance, the denominator is too large and the estimate is attenuated. The numerators are all equal because the part removed from Y lies in the span of the controls and is orthogonal to $\tilde{D}$. Work this out on paper; it is Question 2 of Problem Set 1.

The residuals are also identical, as the theorem promises:

```
all.equal(unname(residuals(lm(y_tilde ~ d_tilde))), unname(residuals(fit)))
#> [1] TRUE
```

7.1 The added-variable plot

```
tibble::tibble(d_tilde, y_tilde) |>
  ggplot(aes(d_tilde, y_tilde)) +
  geom_point(alpha = .12, size = .5, colour = "navy") +
  geom_smooth(method = "lm", se = FALSE, colour = "firebrick", linewidth = .7) +
  labs(x = "Years of schooling | controls", y = "Log earnings | controls") +
  theme_minimal(base_size = 10)
```



Figure 3: Added-variable plot for years of schooling. The slope of the fitted line *is* the multivariate coefficient on `educ`.

Note how much of the spread in `educ` survives residualization ($\text{sd}(\tilde{d}) = 2.71$ versus $\text{sd}(\text{educ}) = 2.82$). When that ratio gets small, the coefficient is being estimated off a thin slice of the data, and its standard error will show it.

8 Omitted variable bias, numerically

Short regression omits verbal ability and parental education; long regression includes them.

```
long <- lm(lnearn ~ educ + exper + I(exper^2) + female + black +
          wordsum + pareduc, data = gss)
short <- lm(lnearn ~ educ + exper + I(exper^2) + female + black, data = gss)

## auxiliary regressions: each omitted variable on the included ones
aux_w <- lm(wordsum ~ educ + exper + I(exper^2) + female + black, data = gss)
aux_p <- lm(pareduc ~ educ + exper + I(exper^2) + female + black, data = gss)

beta_short <- coef(short)["educ"]
beta_long <- coef(long)["educ"]
bias_pred <- coef(long)["wordsum"] * coef(aux_w)["educ"] +
            coef(long)["pareduc"] * coef(aux_p)["educ"]

round(c(short = beta_short,
        long = beta_long,
```

```

    long_plus_bias = beta_long + bias_pred,
    bias = bias_pred), 5)
#>          short.educ          long.educ long_plus_bias.educ          bias.wordsum
#>          0.14053          0.11829          0.14053          0.02225

```

The identity $\tilde{\beta}_1 = \beta_1 + \sum_j \beta_{2j} \delta_{1j}$ holds exactly, to machine precision. Note the direction: ability and family background are positively associated with schooling and positively associated with earnings, so omitting them **overstates** the return to schooling.

Column (1) is the bivariate regression; (2) adds the demographic controls; (3) adds vocabulary score and parental education.

```

modelsummary(list("(1)" = lm(learn ~ educ, data = gss),
                  "(2)" = short,
                  "(3)" = long),
             coef_map = c("educ" = "Years of schooling",
                          "wordsum" = "Vocabulary score",
                          "pareduc" = "Parental education"),
             gof_map = c("nobs", "r.squared"),
             stars = TRUE, output = "markdown")

```

	(1)	(2)	(3)
Years of schooling	0.119*** (0.005)	0.141*** (0.005)	0.118*** (0.006)
Vocabulary score			0.041*** (0.008)
Parental education			0.021*** (0.004)
Num.Obs.	3509	3509	3509
R2	0.129	0.233	0.245
• p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001			

! Important

Everything in this section is *algebra*. It tells you how the estimate moves when you add a control. It does **not** tell you whether the long regression recovers the causal return to schooling, for that you would need to argue that nothing else is omitted, which is a substantive claim, not a statistical one. That argument is what Weeks 5–11 are about.

9 Exercises

Work these during lab; they are not collected, but Problem Set 1 builds directly on them.

1. **The dummy variable trap.** Run `lm(learn ~ factor(region), data = gss)` and look at how many coefficients you get. Now try to include *all nine* region indicators and an intercept. What does `lm()` report, and why? Explain in words: the nine dummies add up to the intercept column, so one of them carries no new information.
2. **The $n - 2$, by simulation.** In the bivariate model, verify that $E[\sum_i \hat{e}_i^2] = \sigma^2(n - 2)$: fix `xs` from the first simulation, generate $y = 8.2 + 0.12x + \epsilon$ with $\epsilon \sim N(0, 4)$, and average $\sum_i \hat{e}_i^2$ over 2,000 replications. Compare to $4 \times (n - 2)$, and to the wrong guess $4 \times n$.
3. **Sign the bias.** Suppose you omit `hours` from the long regression. Predict the sign of the bias in the `educ` coefficient *before* running anything, using the OVB formula and the two auxiliary regressions. Then check.
4. **The FWL standard error.** Compute $s^2 / \sum_i \tilde{d}_i^2$ using `d_tilde` from the FWL section (with s^2 from the full regression), take the square root, and confirm it matches the `educ` standard error printed by `summary(fit)`.
5. **Saturate the key covariate.** Fit `lm(learn ~ factor(educ) + exper + I(exper^2) + female + black, data = gss)`, which gives every year of schooling its own coefficient, and compare it to the linear-in-`educ` version with `anova()`. The F -test you get is exactly the model comparison from *Social Statistics I*: is the reduction in SSE worth the extra parameters?

10 Session information

```
sessionInfo()
#> R version 4.5.2 (2025-10-31)
#> Platform: aarch64-apple-darwin20
#> Running under: macOS Tahoe 26.5.2
#>
#> Matrix products: default
#> BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/Libraries/libBLAS.dylib
#> LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib,
#>
#> locale:
#> [1] C.UTF-8/C.UTF-8/C.UTF-8/C/C.UTF-8/C.UTF-8
#>
#> time zone: America/New_York
#> tzcode source: internal
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods   base
#>
#> other attached packages:
```

```

#> [1] modelsummary_2.6.0 broom_1.0.12      ggplot2_4.0.2      dplyr_1.2.0
#>
#> loaded via a namespace (and not attached):
#> [1] Matrix_1.7-4      gtable_0.3.6      jsonlite_2.0.0
#> [4] compiler_4.5.2    tinytex_0.58      tidyselect_1.2.1
#> [7] tinytable_0.16.0  tidyr_1.3.2       splines_4.5.2
#> [10] scales_1.4.0      yaml_2.3.12       fastmap_1.2.0
#> [13] lattice_0.22-7    R6_2.6.1          labeling_0.4.3
#> [16] tables_0.9.33     generics_0.1.4    lmtest_0.9-40
#> [19] knitr_1.51        backports_1.5.0   checkmate_2.3.4
#> [22] tibble_3.3.1      insight_1.4.6     pillar_1.11.1
#> [25] RColorBrewer_1.1-3 rlang_1.1.7       performance_0.16.0
#> [28] xfun_0.56         parameters_0.28.3 S7_0.2.1
#> [31] datawizard_1.3.0  otel_0.2.0        cli_3.6.5
#> [34] withr_3.0.2       magrittr_2.0.4    mgcv_1.9-3
#> [37] digest_0.6.39     grid_4.5.2        nlme_3.1-168
#> [40] lifecycle_1.0.5   vctrs_0.7.1       evaluate_1.0.5
#> [43] glue_1.8.0        data.table_1.18.2.1 farver_2.1.2
#> [46] zoo_1.8-15        bayestestR_0.17.0 rmarkdown_2.30
#> [49] purrr_1.2.1       tools_4.5.2       pkgconfig_2.0.3
#> [52] htmltools_0.5.9

```